

Chapter 9

Modelling Sustainable Energy Transition for Cities: Case Studies of LEAP, ENPEP-BALANCE, and MoCES



A. Fall and C. A. Mbodji

Abstract The city of Dakar, in Senegal, is endowed with solar photovoltaic, waste-to-energy, and wind energy resources that could support the transition to energy sustainability in the production of electricity for the interconnected grid. Ordinarily, energy planners use modelling software to simulate the integration of this potential into the electricity generation network. In 2018, Ringkjøb et al. identified 75 energy modelling software. The study simulates the integration of the city renewable energy potential in the electricity generation network, using three energy planning software. The main finding is the approach, which is embedded in the software programming interface, affects the model results in terms of competitiveness of renewable energy technologies when compared to existing energy technologies.

Keywords Energy planning · Energy software · System simulation · Computational equilibrium · Agent-based modelling

1 Introduction

Since the 1970s, energy software has been used to describe, plan, and monitor energy systems. The first energy modelling software appeared between the oil crisis of 1973 and the energy crisis of 1979, with the Market Allocation model known as MARKAL (1978) and the Wien Automatic System Planning Package known as WASP III (1979). Since then, the number of energy software has constantly increased, and in 2018, Ringkjøb et al. (2018) identified 75 energy modelling

A. Fall (✉)
Energy Economics Group TU Wien, Vienna, Austria
e-mail: fall@eeg.tuwien.ac.at

C. A. Mbodji
Laboratoire Electronique, Informatique, Télécommunications et Energies Renouvelables,
Gaston Berger University, Saint-Louis, Senegal

software. However, the weight of the model's approach on the energy systems computed is still unclear.

A previous study (Fall, 2020) assessed the potential of renewable energy resources in Dakar and demonstrated a technically achievable potential on solar photovoltaic, waste-to-energy, and wind energy resources. This study uses the figures associated with this potential to explore how different modelling approaches affect energy metrics depending on the software used. The study hypothesized that the modelling approach affects the city energy metrics, and different software can result in different portraits of the future energy system, which, together, affect policy recommendations for transition to sustainability. Most publications on energy modelling software emphasize the modeller's responsibility as a probable cause for this discrepancy; the selection of a modelling software only depending on the user's preferences. However, these preferences do not necessarily translate into an informed decision considering all alternatives, but more than often relates to the accessibility of the modelling software (free download or paid licence), technical knowledge of model operations (inputs and outputs), previous exposure to the software environment (e.g training), etc.

The first objective of the chapter is to introduce MoCES, which approach considers individual agents having rules in planning city energy systems. With this software, the authors bring innovation in the field of energy modelling as agent-based modelling is not frequently used in planning energy systems, despite its popularity in economics. The second objective of the paper is to compare MoCES computation results with two renowned energy planning software that feature different approaches: LEAP and ENPEP-BALANCE. The rationale of this study is to quantify the tractable implications of the modelling approach in planning the transition to energy sustainability for the city.

2 Methodological Approach

The study compares three sample energy software in modelling a future energy system from a single reference energy scenario (2016) and using the same renewable energy technologies. The sample city is Dakar, and figures associated with the city potential on renewable energy capacities (Fall, 2020) are:

- Solar photovoltaic (39.6 MW)
- Waste-to-energy (689.95 GW)
- Wind energy (5951.4 MW)

The sample software in our study has different approaches in modelling future energy systems, which are system simulation for LEAP, computational equilibrium for ENPEP-BALANCE, and agent-based modelling for MoCES. MoCES is an energy planning software designed within the framework of this study.

2.1 Long-Range Energy Alternatives Planning System: LEAP (Stockholm Environment Institute, 2020)

LEAP's approach is a simulation of energy scenarios from a baseline (Current Accounts) to the end of the scenario period (End-Year). The user inputs data of the baseline year in the Current Accounts and includes changes in scenarios in the form of numerical values. In the analysis window (View) of LEAP, the user can input the system energy demand, transformation (conversion from primary energy resources to secondary energy production), and resources potential. The software features a Technology and Environmental Database (TED) that enables the user to connect appliances, devices, fuels, and technologies entered in the model to a database that includes metadata such as pollutant emissions of these appliances, devices, fuels, and technologies. LEAP also provides an option to optimize the energy system modelled.

2.2 Energy and Power Evaluation Programme: ENPEP-BALANCE (Argonne National Laboratory, 2019)

ENPEP-BALANCE's approach is the optimization of an energy network made of energy production (resources), conversion, transport, distribution, and end-use nodes, as well as the flows of energy and fuels among those nodes. Opposite to LEAP and MoCES, the software interface is an empty workspace that the modeller fills in configuring an energy network with embedded nodes and links. ENPEP-BALANCE simultaneously finds the intersection of supply and demand curves for all energy supply sources and all energy uses included in the network. Equilibrium is reached when the model finds a set of market-clearing prices and quantities that satisfy all relevant equations and constraints entered by the user.

2.3 Modelling Cities Energy Systems: MoCES (Fall et al., 2020)

MoCES is an agent-based modelling software that aims at computing the city energy network based on individual agents' energy choices. At city and district levels, MoCES computes the total energy end-use considering the average energy per capita and the population, which is similar to excel-based calculations. The model trades off its basic features in these levels of analysis with a more detailed interface for individual agents, whether it is a residence, commerce, or industrial building.

- On the demand side, MoCES computes the demand of the building energy services, including lighting, cooling, heating, and cooking, while considering these variables: number of plugging appliances and devices in the building, wattage (W), and usage time per day (hours).
- On the supply side, MoCES computes the energy of each production technology accessible in the building and available in the catalogue that includes solar photovoltaic, solar thermal, wind energy, and waste recycling to energy while considering these variables: power (kW), efficiency, capacity factor and operational hours per period (hours). The user can also compute energy from the grid in this window with metering data for benchmark purposes.

Figure 9.1 depicts the MoCES' demand window.

2.4 Data Sources

The data in the reference energy scenario of the model (2016) are from three material sources:

- (a) Report of the Senegal Information Energy System (SIE 2016) that has been collected from the Ministry in charge of Energy (Ministere en charge des Energies, 2018).
- (b) Utility report on the electricity sector in 2016 (SENELEC, 2017) that is accessible online.
- (c) The matrix of data collected during the survey on citizens' energy behaviour performed in different districts of Dakar between November 2018 and May 2019, which was conducted within the framework of the project Sustainable Energy Access for Sustainable Cities (SEA4cities). Annex 1 provides supplementary materials on the survey.

3 Data and Results

From a single reference energy scenario (Scenario 1), which is a simplified—modelled—version of the city energy network in 2016, we model two future energy scenarios (2017–2030) using the sample modelling software:

- Integration of the renewable energy potential in the supply mix (scenario 2)
- Demand-side-management in the residential sector (scenario 3)

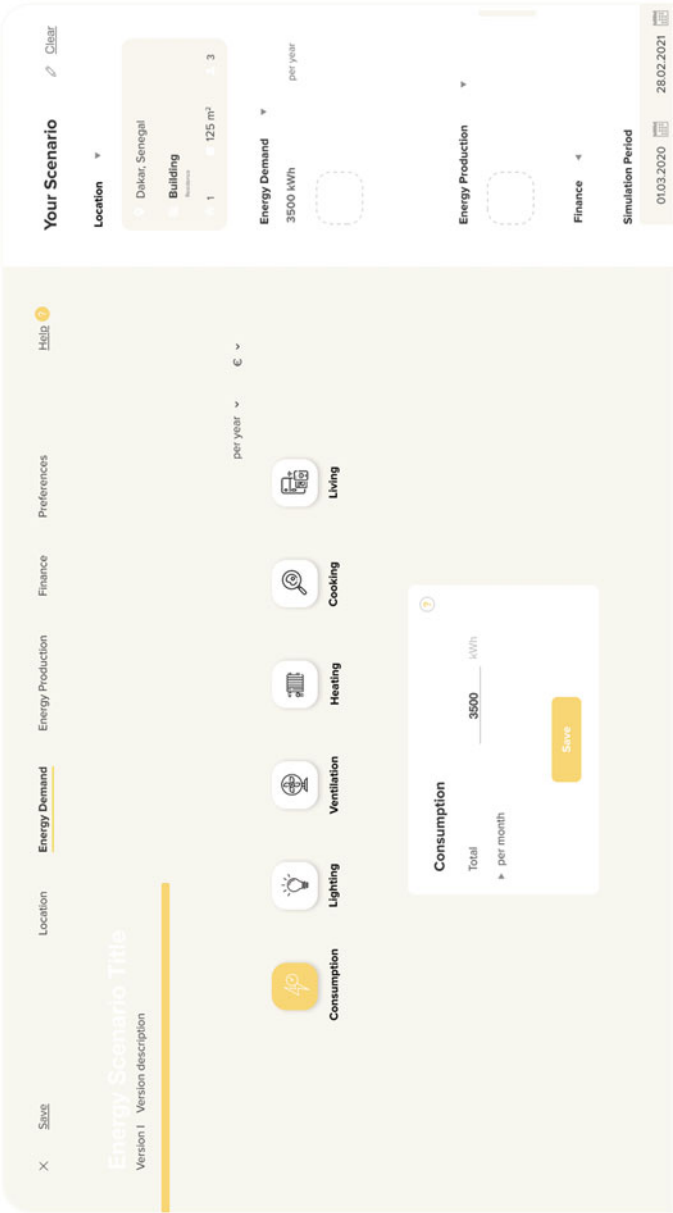


Fig. 9.1 MoCES Demand Window

3.1 Reference Energy Scenario (RES)

Figure 9.2 displays the simplified city energy network in the baseline year (2016). Energy in our network flows from primary energy resources (bottom) to end-use in the residential, commercial and industry sectors (top). We provide detailed information on the network input parameters in the following paragraphs.

3.1.1 Primary Energy Resources

Table 9.1 displays the quantities, conversion factors, and costs in EUR per tonne oil equivalent (toe) of primary energy resources in the network.

The conversion rate USD to EUR is 0.9, which was the average rate in 2016 (Statista, 2020).

Considering reforestation cost of EUR 991 per ha and a charcoal land intensity of 8333.3 kg per ha (Ministere en charge des Energies, 2018), the energy content of the wood species in the study of 20.9 MJ per kg (Fall, 2020), and a wood to charcoal conversion efficiency of 20%, the wood price is 23.8 per metric ton.

Among the primary energy resources in the network, crude oil and coke are imported. Natural gas and wood are domestic resources.

Oil Refining

From imported crude oil (1.1. million tonnes), the local refineries produce diesel, fuel oil, and LPG with an efficiency rate of 88%. The model assumes that any short-fall of diesel, fuel oil and LPG demand in the network is met through imports.

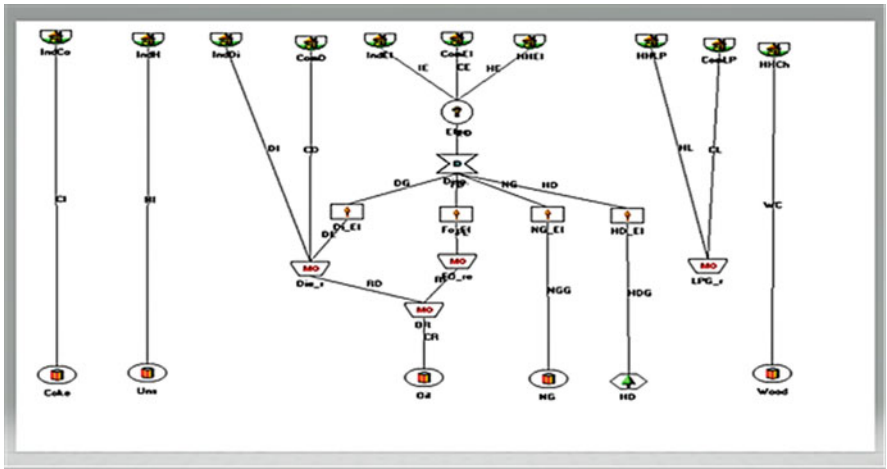


Fig. 9.2 RES City energy network (created with ENPEP)

Table 9.1 RES Primary Energy Resources (2016)

Resources	Consumption (toe) ^a	Price (USD) ^b	Conversion factor (toe)	Price (USD/toe)	Price (EUR/toe) ^c
Coke	416,566	120.59 per tonne	1 tonne = 0.64 toe	187.5	168.75
Crude oil	1,102,202	43.74 per barrel	1 barrel = 0.14 toe	312.43	281.19
Natural gas	416,604	2.87 per million BTU	1 million BTU = 0.026	108.8	97.92
Wood	1691.86	23.8 per tonne	1 tonne = 488 (10–3)	49.79	44.81

^aFigures in the column are from SIE 2016 (Ministere en charge des Energies, 2018)

^bFuel prices are the average of 2016 prices in the international market of commodities downloaded from Statista on February 2020. For wood, we considered the reforestation cost of EUR 991 per ha (Ministere en charge des Energies, 2018)

^cThis column uses information in precedent columns to calculate the unit cost of each primary energy resource in the system per tonne oil equivalent

Table 9.2 Charcoal production parameters

	Km ²	ha
Green areas in the city	161.9	16,190
Green areas with beefwood ^a		81
Charcoal intensity (ha/kg) ^b		0.12
Charcoal potential (kg)		674,583.33
Equivalent wood potential (kg)		3,372,916.67
Equivalent wood energy (MJ)		70,493,958.3
Conversion factor (toe/MJ)		41,868
Equivalent wood production (toe)		1691.86

^a The study assumes beefwood as wood species for the charcoal production. We were not able to find in the literature a specific figure on beefwood share in the woodland of the city. Therefore, we assume 0.5% share (over 200 species available in the woodland). The share could also be understood as a sustainable wood harvesting, where 0.5% of the woodland should be harvested every year

^b The charcoal intensity in kg per ha is from SIE 2016 (Ministere en charge des Energies, 2018)

Charcoal Production

In our simplified energy network, we consider charcoal, because it can be replaced by other fuels in the network, i.e. electricity and LPG in providing the same cooking service. Charcoal comes from the local production of wood that uses kiln with an average efficiency of 20%. Table 9.2 presents information on the wood to charcoal conversion scheme in our model. The model assumes that any shortfall of charcoal demand is met through imports.

3.1.2 Energy Conversion

Electricity Transmission and Distribution

According to the utility (CRSE, 2017), the losses in the electricity transmission and distribution in the interconnected grid were in 2016:

- Transport: 1.2%
- Distribution: 16%

Energy Generation Table 9.3 provides parameters related to secondary energy production (conversion from primary resources) in thousand tonnes oil equivalent (ktoe).

We assume 62% as the share of the city in the interconnected grid generation. This figure is the share of electricity from the interconnected grid consumed in the city in 2016 (ANSD, 2019).

The hydropower resource is outside the city, which geographic location (coastal region at about 20 metres above sea level) does not allow the installation of hydropower systems.

With respect to capacities, the study assumes that the overall installed capacity in the interconnected grid is available to supply the city. Then, we enter in the model

Table 9.3 RES Secondary Energy Production in ktoe

Conversion	Production	Availability rate	Efficiency rate	Cost (EUR/kWh) ^a	Cost (EUR/toe) ^d
Diesel (electricity)	144.3	0.88	0.39	0.07	766.86
Fuel oil (steam)	14.2	0.6	0.39	0.09	1053.85
Natural gas	0.33	0.92	0.39	0.17	1922.81
Hydropower	19.2	0.8	0.8	0.032	371.02
Charcoal	46.7		0.2		123.25
LPG ^b	423.3				65.34
Diesel (end-use) ^c	119.8				494.4

^aThe columns production, availability rate, efficiency rate, and costs of conversion of power plants are from the utility (SENELEC, 2017)

^bThe figures for charcoal and LPG are from SIE 2016. LPG is used in 86% of households at a cost of EUR 0.76 per kg (EUR 4.6 for the 6 kg bottle). The energy intensity of LPG is 0.07 toe per household. Charcoal is used in 14% of households. The energy intensity of charcoal is 0.11 toe per household. Charcoal expense is on average EUR 9.3 per household (Ministere en charge des Energies, 2018)

^cDiesel end-use is a computed figure that reconciles figures of energy end-use in the commercial and industry sectors from SIE (Ministere en charge des Energies, 2018) and our assumptions on the city share of energy consumption from these sectors in the RES. For instance the diesel end-use of commerce is the city commerce end-use estimated as 39.5% of national figures minus other fuels end-use in the sector provided in the SIE report (Ministere en charge des Energies, 2018)

^dThis column converts the unit costs provided in the above-named documents, considering the conversion rate 1 toe = 11,630 kWh

the availability rates provided by the utility for these installations in 2016 and the peak load data per month (Annex 2). The efficiency of thermal units was on average 39% in 2016. The study assumes 80% efficiency for hydropower units.

The dispatching of power plants considers the running cost rule, meaning plants with the lowest generation costs (diesel, fuel oil and steam) are baseload plants, and plants with the highest generation cost (natural gas) are peak load plants.

3.1.3 Energy Demand

Residence

Figure 9.2 shows the energy end-use of households are electricity, charcoal, and LPG. Electricity is used for cooling, lighting, refrigeration, and for the operation of other plugging appliances. Table 9.4 displays the energy intensity of households' electricity services.

For each service, we computed an average of the energy intensity using data from the surveys on Dakar's energy behavior completed in May 2019 (low standard district) and November 2019 (high standard districts). For additional information on the survey, see Annex 1. According to the national statistics agency (ANSD), quoting the utility, the average electricity consumption per household was 1103.9 kWh (ANSD, 2019). Higher figures from the survey data can be explained by the periods of survey, and/or the method used to extrapolate daily consumption averages. For lighting, the average energy intensity per household per year is 409.5 kWh (low standard district) and 602.7 kWh (high standard district). Then the study considers that the city's average is 506.1 kWh per household per year. We use the same method to estimate the energy intensity of other services. The total electricity demand of services in our model amounts to 3066.3 kWh per household per year in the city. Residential grid users in the city were 424,939 in 2016 (CRSE, 2016).

Industry

Total energy consumption of the industry sector was 723 ktoe in 2016 (Ministere en charge des Energies, 2018). Ninety-one (91) per cent of these industries were located in Dakar, according to the 2016 General Survey of Enterprises (ANSD, 2017). Therefore, the energy intensity of industries in our sample city is estimated at 657.9 ktoe (equivalent to 7651.4 GWh), of which 1716.7 GWh of electricity. Other industry energy uses are coke (cement industry), diesel for backup generation and unavoidable steam that results from some industrial (e.g. phosphates) business-as-usual activities.

Table 9.4 Intensity of households' electricity services (in kWh)

Service	Lighting	Cooling	Refrigeration	TVs	Others
	506.1	1153.5	1297.5	73.2	36
Total	3066.3				

Commerce

Among the non-industrial enterprises identified in the 2016 General Survey of Enterprises (ANSD, 2017), 39.5% were located in Dakar. The energy consumption of the commercial sector was 1122 ktoe at the national level (Ministere en charge des Energies, 2018). Therefore, the energy intensity of enterprises in the city is estimated at 443.2 ktoe. Other energy uses of the commerce sector are diesel that fuels backup generators, and LPG for some businesses (e.g. restaurants).

3.2 Renewables in Electricity Generation (Scenario 2)

In scenario 2, we assume:

- (a) City population grows at the rate of 3.7% per year; figure provided by the World Bank for Dakar during the period 1990–2018 (World Bank Group, 2019).
- (b) Number of households connected to the grid increases by 4.4%, which is an average of the period 2009–2016 computed from the annual utility reports of the period.
- (c) The city's renewable energy potential (Fall, 2020) is integrated into the electricity generation mix.

Table 9.5 summarizes figures related to the city's renewable energy potential (Fall, 2020).

Bioenergy potential of the city for electricity generation is made of waste-to-energy. Technologies for conversion of waste to electricity are anaerobic digestion (AD) and dendro liquid energy (DLE).

3.3 Demand-Side-Management in the Residence Sector (Scenario 3)

In scenario 3, the study assumes from the Scenario 2 as baseline, improvements in households' energy behaviour compared to observations during the survey:

Table 9.5 Renewables' potential capacity and costs

Technology	Bioenergy		Solar PV	Onshore Wind
	DLE	AD		
Capacity (MW)	33.6	6	689,945.3	5951.4
Efficiency factor (%)	80	66	21	40
Capacity factor (%)	84	84	17	31
Installation cost (EUR/kW)	517	1890	1609	1609
O&M (EUR/kWh)	0.01	0	0.027	0.01
CO ₂ cost (EUR/kWh)	0	0.01		

- (a) Lighting energy intensity decreases by 2/3, equivalent to a retrofit of bulbs from halogen (18 Watt) to compact fluorescent light (CFL) (6 W) standard or from CFL to LED (2 W) standard at the constant brightness of 200 lumens per m².
- (b) Cooling energy intensity decreases by 20% equivalent to an increase of the balance temperature point (comfort temperature) by 1degree Celsius. For example a building with constant environmental factors such as air exchange factor, specific heat capacity, and air density will require 164 Wh energy to reach 22 degree Celsius balance temperature point when the outside temperature is 25 degree Celsius. When we increase the balance temperature point to 23 degree Celsius, the cooling energy requirement becomes 131 Wh, meaning a decrease by 20%.
- (c) Refrigeration energy intensity decreases by 32% equivalent to retrofit from low standard fridges of 220 Watt to medium standard fridges of 150 Watt.
- (d) Energy intensity of TVs, phone, and other appliances decreases by 10% due to manufacture improvement in battery autonomy or sleep mode consumption, without additional action from the user.
- (e) All households in the city use LPG for cooking.

4 Discussion of Results

4.1 Reference Energy Scenario (RES)

Figure 9.3 displays the city Reference Energy Scenario (2016) generated with LEAP.

Legend: The energy balance is presented in tonnes oil equivalent, with all input parameters converted with the embedded LEAP units' converter. It is possible to convert the figures to other energy units such as GWh for electricity, using the international Energy Agency online unit converter accessible at <https://www.iea.org/reports/unit-converter-and-glossary>.

The overall electricity generation in the network was 697.5 ktoe in 2016. This generation mainly relies on diesel (85.8%), and fuel oil (8.5%) produced from crude oil imported by refineries. Hydropower (5.5%) and natural gas (0.2%) complete the electricity generation resources. Energy losses (617.4 ktoe) include refineries, power plants, and network losses. It represents more than the electricity distributed to end-users due to the relatively low efficiency of thermal generation units (diesel, fuel oil and natural gas). As electricity demand for end-use sectors (280.1 ktoe) is more than the system's supply capacities (229 ktoe) after transmission and distribution losses, the network imports 51.1 ktoe electricity to meet the demand of end-use sectors. The heat use of industry comes from domestic resources (natural gas), and there is no import of heat in the RES, despite the fact LEAP displays it with 0 as value.

4.2 Renewables in Electricity Generation (Scenario 2)

4.2.1 Leap

Figure 9.4 displays the city energy network under scenario 2 in LEAP.

Electricity generation increased by 179% and supplies the overall city electricity demand (imports = 0). Wind energy resources converted to electricity at the average cost of EUR 8.3 cents per kWh supplies 98.8% of the city electricity demand. This generation cost is higher than the diesel unit cost (EUR 7 cents per kWh) and hydropower unit cost (EUR 3 cents per kWh). However, both technologies run with imported fuel, and LEAP computes results, prioritizing domestic resources per default. In this scenario, crude oil imports decreased by 52.1%, and its refining only produces LPG and diesel for the commerce and industry sectors end-uses.

The remaining electricity generation (1.2%) is from waste-to-energy (anaerobic digestion and dendro liquid energy), which is the cheapest domestic resource. The System cost is EUR 5864.6 mio.

4.2.2 ENPEP-Balance

Figure 9.5 displays the city energy network under scenario 2 with ENPEP-BALANCE.

Legend: The model constant parameters are:

- Premium multiplier that we assumed at 1 to indicate neutrality over fuels available as options. Premium multipliers reflect the preference for a fuel over others. A multiplier greater than 1 raises the price of competing energy products in the market share equation. A multiplier lower than 1 lowers the price of competing energy products in the market share equation.
- Cost sensitivity factor that we assumed at 0.5. The 0 value is an extreme case and indicates the least degree of the fuel share sensitivity to prices. A value above 1 indicates a higher degree of the fuel share sensitivity to relative prices.
- Lag factor that we assumed at 1. The lag value ranges between 0 and 1. A value of 1 indicates there is no lag, and shares respond to current prices. A value of 0 indicates no response to prices, meaning base-year shares are maintained throughout the study period.

In ENPEP, the share of a fuel in the city supply mix is inversely proportional to its cost. The market share of the fuel is its relative price over the sum of all fuels' relative prices. Therefore, the ENPEP model returns a situation where all available fuels are in the supply mix; the cheapest option having the higher share. In 2017, DLE was the cheapest option (EUR 1.9 cents per kWh), but it has a capacity constraint (33.6 MW) that limits generation to 17.1 thousand tonnes oil equivalent per year. The other waste-to-energy technology (AD) was also used at full capacity (6 MW) as the third-cheapest generation option. In between, hydropower import

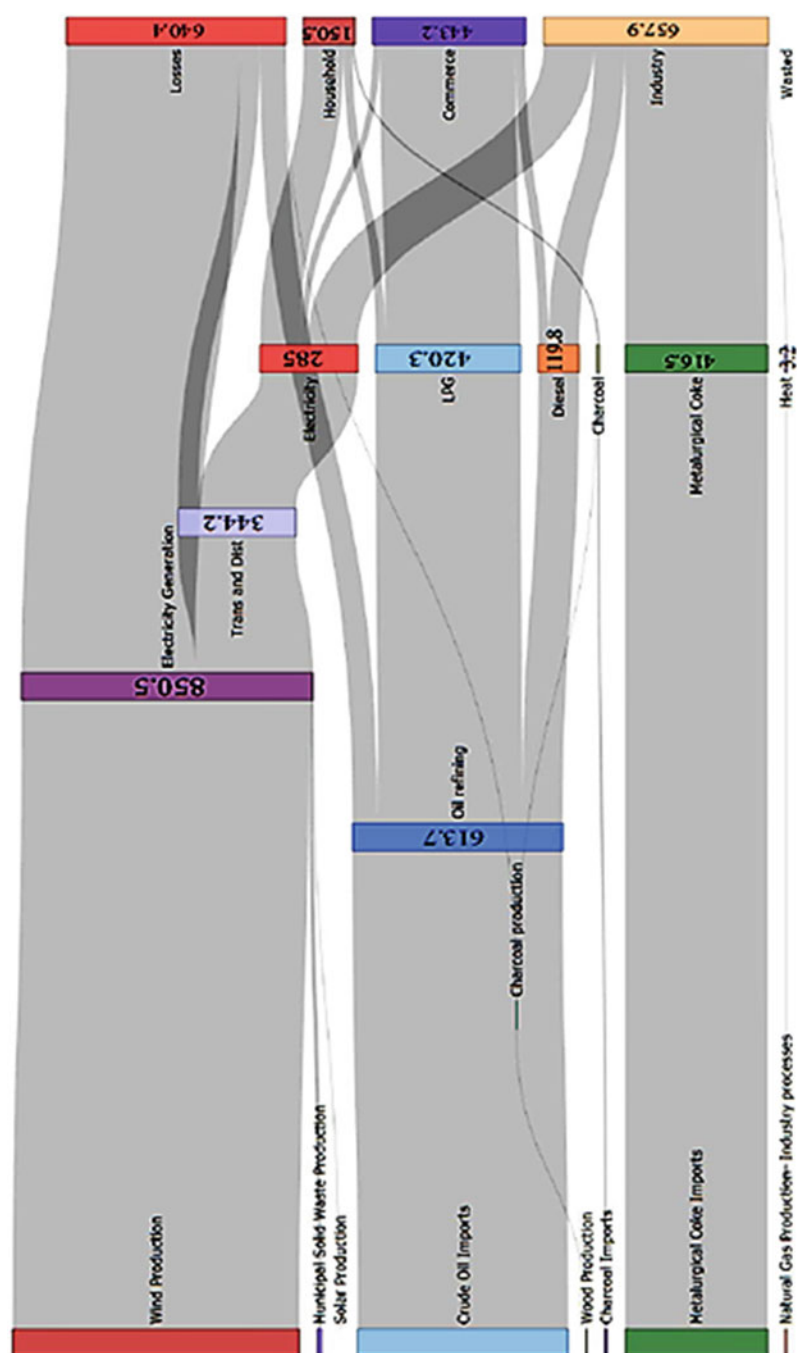


Fig. 9.4 LEAP-2017 City Energy System (in ktce)

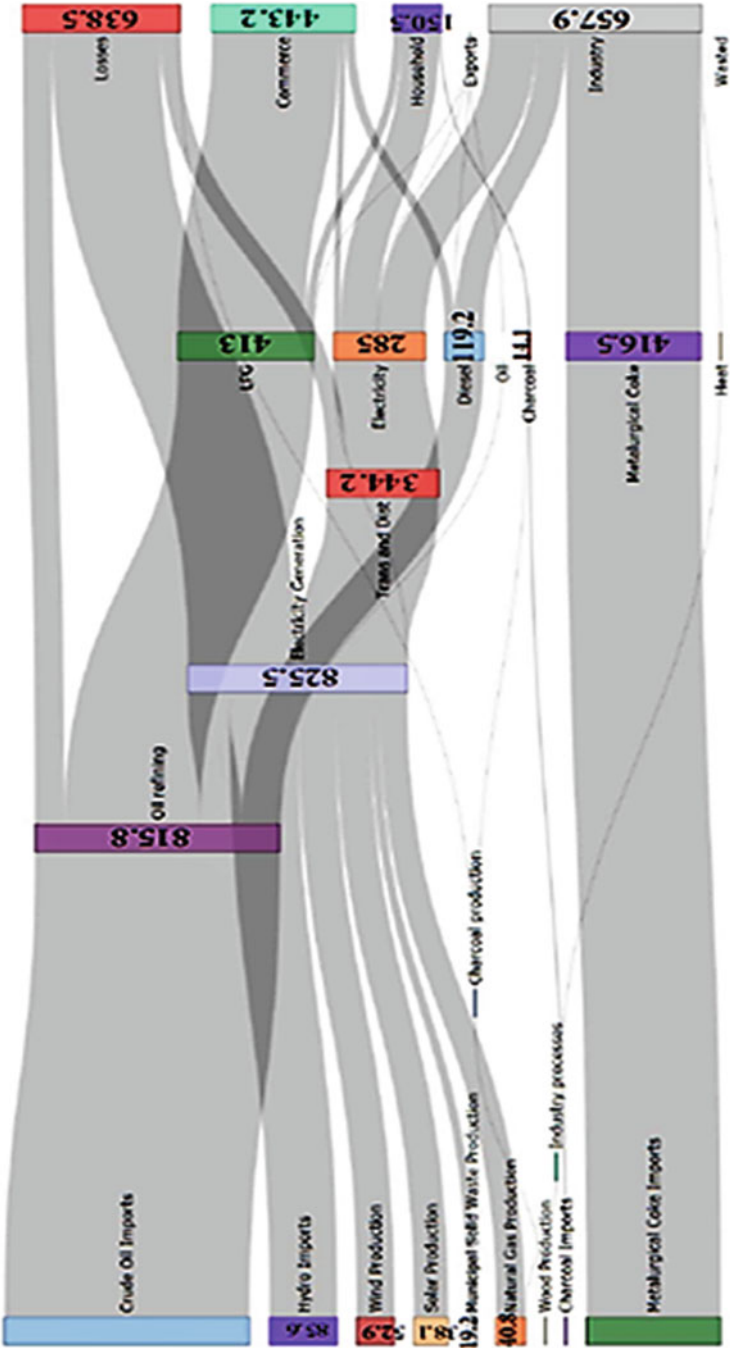


Fig. 9.5 ENPEP-2017 City Energy System (in ktoe)

with a unit generation cost of EUR 3.2 cents per kWh is the main supplier of the grid, followed by wind energy. Diesel generation decreases by 74% compared to the RES. Natural gas with the highest generation cost (EUR 17 cents per kWh) has the lowest contribution to grid supply. The overall system cost is EUR 943.2 mio.

4.2.3 MoCES

Figure 9.6 displays the future energy system under scenario 2 in MoCES.

Similar to ENPEP, the full potential of municipal solid waste-to-energy is integrated, as being the first and third-cheapest electricity generation options in the mix. It is followed by hydropower, diesel, and wind. Hydropower import enters the generation mix limited by its baseline import capacity, and because it is possible to limit power capacity (kW) in the software production window. About 11% of the city's wind energy potential is used to complete the mix. System cost is EUR 1088.8 mio, which is higher than the ENPEP-BALANCE resulting system cost.

4.3 *Demand-Side-Management in the Residence Sector (Scenario 3)*

Figure 9.7 displays result in LEAP of Scenario 3 that assumes, from Scenario 2, an improvement of the households' energy behaviour in terms of energy demand for cooling, lighting, refrigeration, cooking, and other domestic services.

4.3.1 LEAP

Electricity demand represents the main part of a household's energy demand, but it only increased by an average 1.52% during the period due to improvements in the energy performance of electric appliances. In comparison, LPG use increased by an average 5.5% during the period to account for progressive (interpolate function) replacement of charcoal and increase of the number of the city's households. Charcoal demand decreases by an average 19.8% per year to reach zero by 2030.

4.3.2 ENPEP-Balance

- Electricity demand increases by an average 2% over the 15-year period.
- LPG demand increases by an average 1.25%, as well as charcoal demand, because the presence in the energy mix of a fuel depends on its price, consequently charcoal quantities cannot reach zero as long as the fuel is priced. The

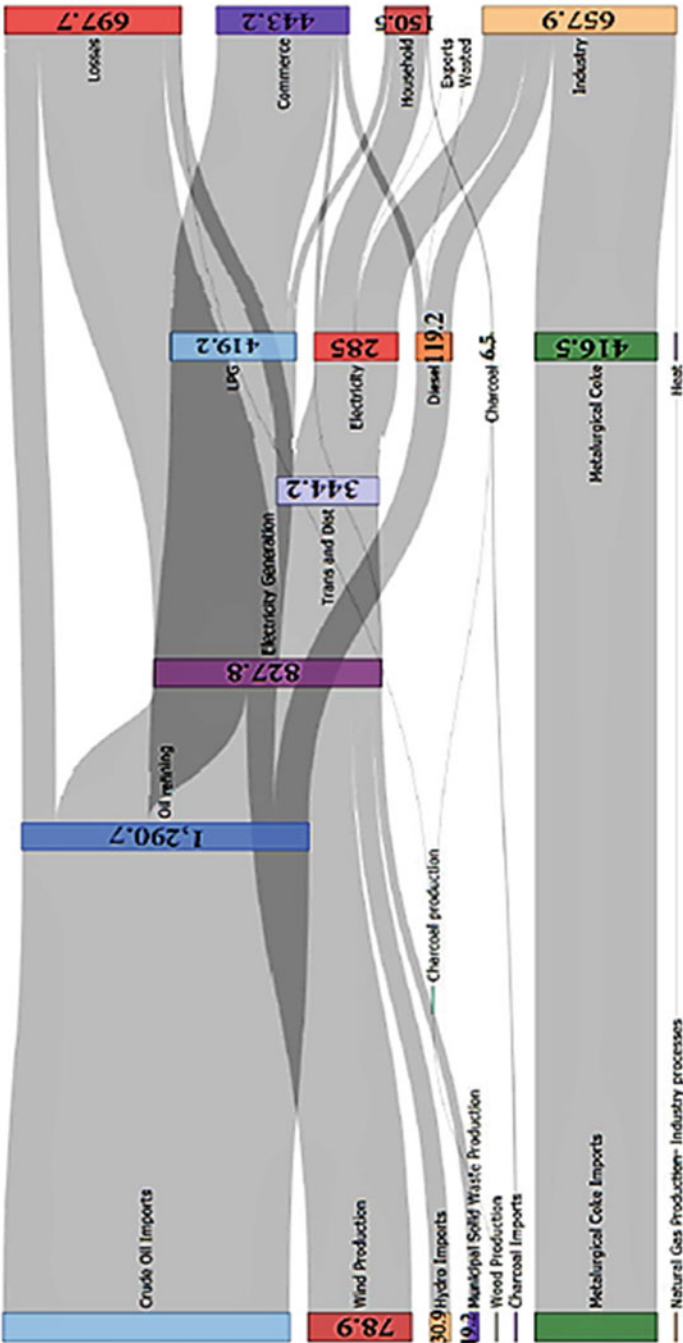


Fig. 9.6 MoCES-2017 City Energy System (in ktOE)

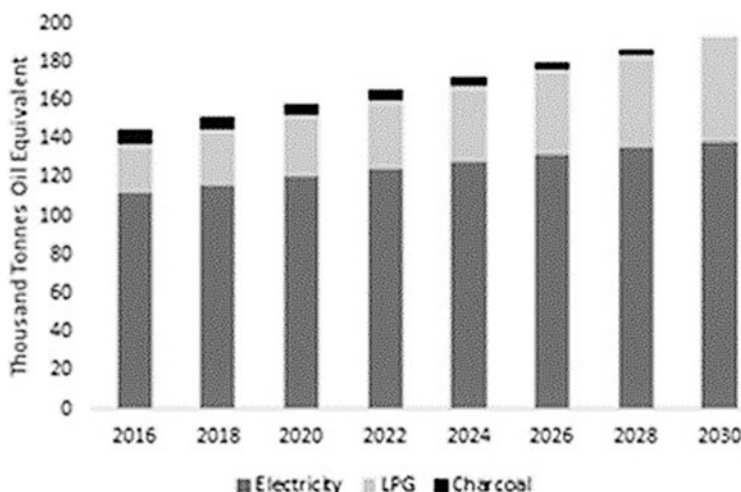


Fig. 9.7 Energy demand per fuel in the residence sector

growth of charcoal demand is driven by the increasing number of households, as for other fuels.

- Gross electricity generation, before losses, decreases by an average 1.4% due to decreasing demand compared to scenario 2.

As the model always runs to an equilibrium, all electricity generation technologies decrease generation quantities in the proportion of their market shares.

4.3.3 MoCES

- Electricity demand decreases by an average 0.17% over the 15-year period.
- LPG demand increases by an average 3.3%, while the charcoal demand decreases by an average 23.3% to reach zero by 2030.
- Gross electricity generation, before losses, decreases by 1.5% due to a decreasing demand compared to Scenario 2.

Compared to Scenario 2, the demand-side-management decreases the wind energy generation by 48% equivalent to a 572 MW wind power plant or the decommissioning of 310 MW diesel capacity by 2030.

In the following paragraphs, we discuss the four main findings from the models.

The selection of the modelling software affects the future energy system. From a single reference energy scenario, we derived different results regarding supply mix and system cost. Therefore, the claim that the outcomes of an energy planning model only depends on subjective considerations (the modeller preference) do not hold, as objective factors like the approach and the algorithm formulae also play out in the outcomes.

One hundred (100) per cent renewables supply mix such as in Scenario-2 LEAP does not guarantee the lowest system cost. Energy transition in the conditions of Scenario 2-LEAP is the most expensive for the city.

Dispatch of the city's generation technologies are considered to be the running cost rule, meaning technologies with the lowest generation cost (long-run generation cost) enter first the supply mix but, this does not guarantee the lowest electricity production cost. The ENPEP-BALANCE scenario-2 that integrates a share of all accessible technologies (with renewable and non-renewable energy resources) has the lowest cost compared to Scenario 2-LEAP (waste-to-energy and wind) and Scenario 2- MoCES (diesel, hydropower, waste-to-energy, and wind).

Accessible demand-side-management in the residence sector has a significant impact in the system's generation quantities. Measures such as retrofitting lamps, setting higher balance temperature point (BTP) in cooling can save up to the equivalent of 572 MW wind energy. The adoption of 100% LPG cooking saves 17 thousand tonnes oil equivalent of wood, which corresponds to a woodland area of 41 km².

5 Conclusion

The study demonstrates that energy modelling software can integrate different dimensions of transition to energy sustainability at the city level, including the selection of electricity generation technologies, security of supply, and improved efficiency in energy use. Moreover, it particularly shows that different pathways exist to reach the same objective of transition to energy sustainability in the city network, and each produces different externalities. LEAP produces the most secure future energy scenario for the city by only using domestically available resources, but it is also the most expensive option. ENPEP displays the cheapest future energy scenario for the city, but it is also a less secure option as it continues to rely on imported sources that are cheaper than domestically available resources. MoCES displays results somewhere between security (more wind in supply) and affordability (presence of diesel).

Still, existent software also features limitations. These limitations include the abstraction of relevant energy parameters that affect the future system modelled by the software and the absence of flexibility in integrating different (agents) rationales on energy demand and supply options. Energy is a social good, as individual agents produce and consume its services; therefore, a relevant modelling approach should integrate the complexity of these agents' rationales to produce and/or to consume it. The cities' efforts to achieve energy sustainability (SDG-7) and urban liveability (SDG-11) by 2030 require relevant accounting methodologies and consistent model metrics (Grubler et al., 2012). MoCES, with an agent-based modelling approach, is a contribution to addressing both the relevance and consistency issues in energy planning models.

A. Annexes

Annex 1- Supplementary materials related to the SEA4cities survey on energy behavior of Dakar citizens.

The Table 9.6 summarizes the survey sample data.

Table 9.6 Survey strata

District	Residence buildings (a)	Sample size (b)	Number of residences surveyed (c)
Diamniadio	2165	327	368
Fann-point E Amitie	2128	326	360
Total	4293		728

^aThe number of residence buildings in a district is from ANSD (Diamniadio) and from the municipality of the district (Fann-Point E—Amitie)

^bThe sample size is computed with a confidence interval of 95% and $p = 0.5$

^cThe number of residence buildings that completed the survey questionnaire

Table 9.7 Average energy consumption

	Diamniadio	Fann—Point E—Amitie	City (Average)
Lighting (Wh) per day ^(a)	369,087.5	403,478.5	
Number of days (2016)	366	366	
Number of households	330	245	
Lighting (kWh per household)	409.4	602.7	506.1
Cooling (kWh per household) ^(b)	995.8	1311.2	1153.5
Refrigeration (kWh per household) ^(c)	1041.7	1553.3	1297.5
TV (kWh per household) ^(d)	35.8	110.7	73.2
Other appliances (kWh per household) ^(e)	29.7	42.4	36
Total (kWh per household)	2512.3	3620.3	3066.3
Total (toe per household)			0.26

^aThe row summarizes the total lighting energy of surveyed buildings per day. Lighting per household per day is computed with the following parameters from the questionnaire: type of lamps, lamp wattage, number of lamps in the building, and hours of operation per day

^bThe row is an average of energy consumed by space cooling appliances present in the buildings surveyed. The space cooling appliances from the completed questionnaires are air conditioners, ceiling and standing fans. Cooling energy per appliance per day is computed with the following parameters: appliance type, power, and average hours of operation per day in dry and rainy seasons

^cThe row is an average of energy consumed by fridges present in the buildings surveyed. Refrigeration energy is computed with following parameters: type of fridge, power, and hours of operation per day assumed to 24 hours for all fridges

^dThe row is an average of energy consumed by TVs present in the buildings surveyed. TV energy is computed with following parameters: type of TV, power, and hours of operation per day
Other appliances tracked in the survey are phones and laptops. The energy per appliance is computed with following parameters: number of appliances in the building, power and recharge hours per day (average value provided by the respondent)

Table 9.8 Grid Peak load (MW)

Month	Senegal ^(a)	Dakar ^(b)	in % of Dakar Peak
January	474	293.88	84.6%
February	476	295.12	85.0%
March	477	295.74	85.2%
April	489	303.18	87.3%
May	495	306.9	88.4%
June	528	327.36	94.3%
July	532	329.84	95.0%
August	544	337.28	97.1%
September	547	339.14	97.7%
October	560	347.2	100.0%
November	531	329.22	94.8%
December	512	317.44	91.4%

^aPeak load data at the country level in 2016 is from the utility (SENELEC, 2017)

^bPeak load data for the city assumes 62% of the country peak load. This figure was the share of electricity from the interconnected grid consumed in the city in 2016 (ANSD, 2018)

Dates of Survey

The survey in Diamniadio was completed between April and May 2019 (Dry season with average temperature of 30 degree Celsius).

The survey in Fann-Point E—Amitie was completed during November 2019 (End of the rainy season with average temperature of 25 degree Celsius). The Table 9.7 compiles results of the excel-based analysis of data collected during the survey.

Annex 2- Peak load data of the interconnected grid in 2016 (SENELEC, 2017).

Table 9.8 provides peak load data in the interconnected grid for 2016.

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